UNIT 1

Introduction to Quantum Theory and Technologies

Unit 1 at a glance

1. Classical → quantum transition 2. Superposition, entanglement, uncertainty, wave-particle duality

3. Classical vs quantum mechanics 4. States & measurement 5. Electrons, photons, atoms

6. Quantization 7. Why quantum? 8. Quantum computing/communication/sensing 9. National quantum missions

1. From Classical to Quantum Physics

Classical physics (Newton, Maxwell, Boltzmann) explains the everyday world: falling apples, planets, heat engines, light as waves. By the late 1800s many physicists felt physics was 'almost complete'. But a few small puzzles refused to fit - and they cracked the whole picture open.

» The puzzles classical physics could not solve

- *Black-body radiation: a hot object's colour/spectrum. Classical theory predicted infinite energy at short wavelengths (the 'ultraviolet catastrophe'). Wrong!*
- *Photoelectric effect: light knocks electrons out of metal only if its frequency is high enough - however bright the light is. Waves could not explain this.*
- *Atomic stability & spectra: why do atoms emit light only at sharp, fixed colours (lines)? Why don't electrons spiral into the nucleus?*
- *Specific heat of solids at low temperature also disagreed with classical predictions.*

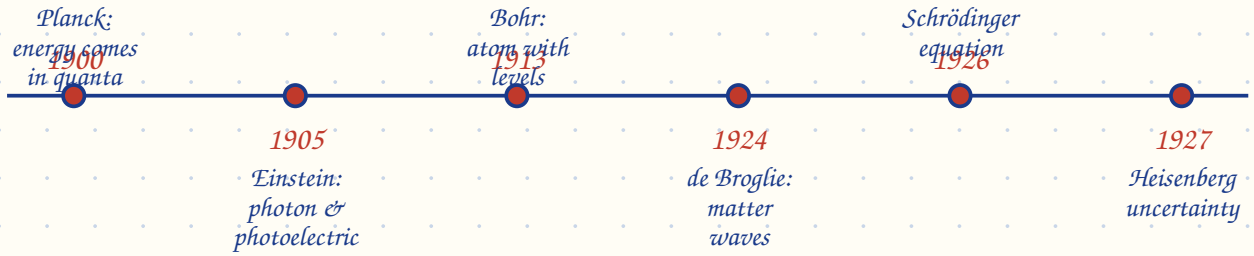


Fig 1: Key milestones in the birth of quantum theory

» The big ideas that fixed it

- *Max Planck (1900): energy is emitted/absorbed in tiny packets called quanta, $E = h\nu$, where $h = 6.626 \times 10^{-34}$ J·s is Planck's constant.*
- *Albert Einstein (1905): light itself is made of packets - photons, each with energy $E = h\nu$. Explained the photoelectric effect (Nobel Prize 1921).*

- *Niels Bohr (1913)*: electrons in atoms occupy only certain allowed energy levels; they jump between them by emitting/absorbing photons.
- *Louis de Broglie (1924)*: every particle has a wavelength $\lambda = h/p$ (matter waves).
- *Schrödinger, Heisenberg, Born, Dirac (1925-28)*: a complete mathematical theory - quantum mechanics.

> ★ *Remember*
> Quantum mechanics is the theory of the very small (atoms, electrons, photons) where energy comes in discrete packets and nature is described by probabilities, not certainties.

2. Fundamental Principles (Conceptual)

» 2.1 Wave-Particle Duality

Light and matter behave sometimes like *waves* (spreading, interfering) and sometimes like *particles* (arriving in lumps at a point). Which behaviour you see depends on how you set up the experiment. This is not a contradiction - quantum objects are simply a new kind of thing that is neither a classical wave nor a classical particle.

- *Wave evidence*: interference and diffraction (Young's double-slit with light; also seen with electrons, atoms, even large molecules).
- *Particle evidence*: photoelectric effect, Compton scattering, single dots on a detector screen.
- *Double-slit experiment*: send electrons one at a time through two slits. Each lands as one dot (particle), yet after thousands of dots an interference pattern (wave) appears. If you detect which slit each electron used, the pattern vanishes!



Fig 2: Double-slit experiment - waves interfere, detection makes dots

> *de Broglie wavelength*
> $\lambda = h / p = h / (m\,v)$
> Bigger momentum → smaller wavelength. A cricket ball's λ is ~10^{-34} m (unobservable) but an electron's λ is comparable to atom size - so electrons show wave effects. This is why quantum effects are 'hidden' in daily life.

» 2.2 Superposition

A quantum system can exist in a combination of several states at once. Example: an electron's spin can be 'up' AND 'down' simultaneously, written $|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$ where $|\alpha|^2$ and $|\beta|^2$ are the probabilities of finding up or down when measured, and $|\alpha|^2 + |\beta|^2 = 1$.

- It is NOT that we are ignorant of which state it is - the system genuinely has no single definite value yet.
- Superposition is what makes interference possible and is the raw resource for quantum computing (qubits).
- Schrödinger's cat (thought experiment): a cat linked to a radioactive atom is 'alive + dead' until observed. Used to point out how strange superposition is when scaled up.

» 2.3 Entanglement

Two or more particles can be linked so that their properties are correlated no matter how far apart they are. Measuring one instantly tells you about the other. Einstein called it 'spooky action at a distance'. Example: two photons prepared so that their polarizations are always opposite; each alone is random, but together they are perfectly correlated.

- The whole system has a definite state, but the individual parts do not.
- Entanglement cannot be used to send messages faster than light (the individual results are random) - no-signalling.
- Bell's theorem + experiments (Aspect, Clauser, Zeilinger - Nobel 2022) showed nature really is non-local; no 'hidden variable' classical explanation works.
- Entanglement = key resource for quantum teleportation, quantum cryptography and quantum computing speed-ups.

» 2.4 Heisenberg's Uncertainty Principle

Statement

$\Delta x \cdot \Delta p \geq \hbar/2$ ($\hbar = h/2\pi$)

Also: $\Delta E \cdot \Delta t \geq \hbar/2$

You cannot know both the position and momentum of a particle with perfect precision at the same time. Squeezing one makes the other fuzzier.

- It is a basic property of nature (wave-like behaviour), NOT merely a flaw of our instruments.
- Position-momentum, energy-time, and spin components along different axes are such 'incompatible' pairs.
- Consequences: atoms are stable (electron can't collapse into the nucleus), zero-point energy exists, and quantum cryptography is secure (measuring disturbs the state).

Exam tip / Remember

The four pillars to learn by heart: Superposition, Entanglement, Uncertainty, Duality. Give one definition + one example for each - this is a common 5-mark question.

3. Classical vs Quantum Mechanics - Theoretical Comparison

Feature	Classical mechanics	Quantum mechanics
Scale	Everyday, large objects; planets	Atoms, electrons, photons, small scales
State described by	Position & momentum (definite values)	Wavefunction / state vector $\|\psi\rangle$
Predictions	Deterministic: same start → same result	Probabilistic: only probabilities of outcomes
Energy	Continuous - any value	Often discrete (quantized)
Measurement	Does not disturb the system (ideal)	Measurement changes the state (collapse)
Nature of matter	Particles OR waves	Wave-particle duality
Combination of states	Object is in one state at a time	Superposition of many states
Correlations	Local; due to common cause	Entanglement; non-local correlations
Equation	Newton: $F = ma$	Schrödinger: $i\hbar\, \partial\psi/\partial t = \hat{H}\psi$
Limit	—	Reduces to classical when action $\gg \hbar$ (correspondence principle)

Correspondence Principle (Bohr)

For large systems/quantum numbers, quantum predictions merge smoothly into classical ones. So quantum theory does not 'throw away' Newton - it contains it as a special limit.

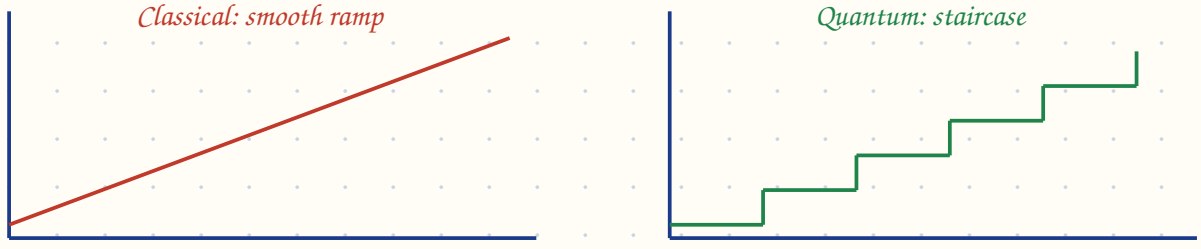


Fig 3: Classical = continuous ramp; quantum = discrete staircase

4. Quantum States and Measurement

» 4.1 What is a quantum state?

A quantum state is the complete description of a quantum system, written as a vector $|\psi\rangle$ (Dirac 'ket' notation). It carries all the information that can be known. For a two-level system: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are called probability amplitudes (complex numbers).

- Born rule: probability of outcome = $|\text{amplitude}|^2$.
- Amplitudes can add or cancel (interference) - probabilities from classical chance cannot cancel, which is the big difference.
- The state evolves smoothly and deterministically between measurements (Schrödinger equation).

» 4.2 The nature of observation (measurement)

- Before measurement: system may be in a superposition.
- During measurement: we interact with the system using a device and get ONE definite result (e.g. 'up').
- After measurement: the state is now the one we saw - this sudden change is called collapse of the wavefunction.
- Which result we get is random; only the probabilities are predictable. Repeating on identical systems gives a statistics.

Measurement - key points

- Measuring is an active process, not passive 'looking'.
- Measuring one property can destroy information about an incompatible one (uncertainty principle).
- 'Observer' does NOT need to be a human - any interaction that records the information acts as a measurement.
- Different interpretations exist (Copenhagen, Many-Worlds, Pilot-wave, Decoherence) - they differ in explanation, not in predictions.

- No-cloning theorem: an unknown quantum state cannot be copied perfectly. This protects quantum communication from eavesdropping.
- Decoherence: interaction with the environment effectively 'measures' the system, turning superposition into ordinary mixtures - why large objects look classical.

5. Overview of Quantum Systems

System	Quantum properties used	Where it is used
Electron	Spin (↑/↓), charge, energy levels; shows interference	Quantum dots, spin qubits, electron microscopes, semiconductors, transistors
Photon	Polarization (H/V), path, time-bin; travels fast, weakly interacts	Quantum communication, QKD, photonic computing, quantum optics
Atom / Ion	Discrete electronic energy levels, nuclear spin	Atomic clocks, trapped-ion quantum computers, laser cooling, sensors

» Electrons

Electrons are charged particles with an intrinsic property called spin (spin-½): it has two measurable values, 'up' and 'down' along any chosen axis. Electrons in atoms and solids occupy discrete energy states and obey the Pauli exclusion principle (no two electrons in the same state), which gives the periodic table and the structure of matter.

» Photons

A photon is a quantum of light: massless, travels at c, energy $E = h\nu = hc/\lambda$, momentum $p = h/\lambda$. Its polarization (direction of the electric field) gives a natural two-level system (horizontal/vertical). Photons barely interact with the environment, so they carry quantum information over long distances through optical fibre or space.

» Atoms

Atoms have electrons in discrete orbitals; transitions between levels absorb/emit photons of exact frequency, giving spectral lines. This exactness is exploited in atomic clocks (the SI second is defined from caesium-133 transitions at 9,192,631,770 Hz). Cooled, trapped atoms/ions are among the best qubits and sensors.

6. The Concept of Quantization

Quantization means that certain physical quantities can take only specific discrete values instead of any value in a continuous range. Energy of bound systems (atom, molecule, particle in a box, harmonic oscillator), angular momentum, spin and charge are all quantized.

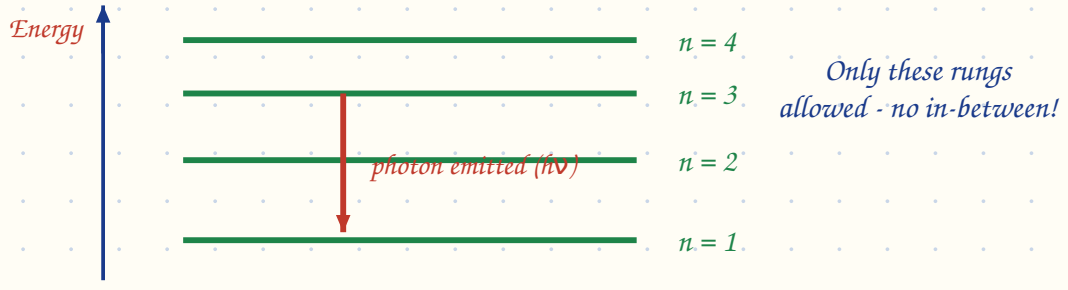


Fig 4: Discrete energy levels of an atom; a jump emits a photon

Energy levels - formulas

Hydrogen atom (Bohr): $E_n = -13.6\ eV / n^2$ ($n = 1, 2, 3 \ldots$)

Photon emitted/absorbed in a jump: $h\nu = E_{high} - E_{low}$

Particle in a 1-D box: $E_n = n^2 h^2 / 8mL^2$

Quantum harmonic oscillator: $E_n = (n + \frac{1}{2})\ \hbar\omega$ - note the non-zero ground state (zero-point energy).

- *Why discrete? Confinement of a wave forces it to fit like a standing wave - only certain wavelengths (hence energies) fit, like guitar-string harmonics.*
- *Ground state = lowest energy; excited states = higher; atoms relax by emitting a photon.*
- *Quantized levels let us build qubits: choose two levels as $|0\rangle$ and $|1\rangle$.*
- *Applications: lasers, LEDs, spectroscopy, solar cells, atomic clocks, MRI.*

Exam tip / Remember

Be ready to derive/quote: $E = h\nu$, $\lambda = h/p$, $\Delta x \Delta p \geq \hbar/2$, $E_n = -13.6/n^2\ eV$. Also know: $h = 6.626 \times 10^{-34}\ J{\cdot}s$, $1\ eV = 1.6 \times 10^{-19}\ J$.

7. Why Quantum? Strategic, Scientific & Technological Significance

» Scientific significance

- *Foundation of modern science: chemistry, materials, condensed-matter, nuclear and particle physics, cosmology all rest on quantum theory.*
- *Tests the nature of reality: non-locality, measurement, information.*
- *New simulation tool: Feynman (1981) - 'nature isn't classical, so to simulate nature use a quantum computer'.*

» Technological significance

- *First quantum revolution (1900-1980s): we learned the rules and built technologies using them - transistor, laser, LED, MRI, GPS clocks, solar cells. The digital world runs on it.*
- *Second quantum revolution (1990s-now): we now deliberately control individual quantum systems (single photons, atoms, qubits) and use superposition and entanglement as resources → quantum computers, quantum networks, quantum sensors.*

» Strategic significance

- Security: a large quantum computer running Shor's algorithm can break RSA/ECC encryption → need for post-quantum and quantum-safe cryptography ('harvest now, decrypt later' threat).
- Economy: expected to add hundreds of billions of dollars of value in pharma, finance, logistics and materials over the coming decades.
- Defence & navigation: GPS-free quantum navigation, stealth detection, secure military communication.
- Technological sovereignty: nations race to own the hardware, talent and supply chains - hence national quantum missions.

1st quantum revolution	2nd quantum revolution
Uses quantum LAWS in bulk materials	Controls INDIVIDUAL quantum states
Transistor, laser, LED, MRI, atomic clock	Qubits, QKD, quantum sensors, quantum internet
We use the effects	We engineer superposition & entanglement

8. A Snapshot of Quantum Technologies



Fig 5: The three pillars of quantum technology

» Quantum Computing

Uses qubits in superposition and entanglement to process information in fundamentally new ways. Promising for factoring (Shor), unstructured search (Grover), simulating molecules and materials, and certain optimization problems. Hardware: superconducting circuits, trapped ions, photons, neutral atoms, spins in silicon. Currently in the NISQ (noisy, intermediate-scale) era with 50 - 1000+ noisy qubits.

» Quantum Communication

Transmits quantum states (mostly photons). Main uses: Quantum Key Distribution (QKD) for provably secure keys (BB84, E91), quantum teleportation, and eventually a quantum internet linking quantum computers. Any eavesdropping disturbs the photons and is detected.

» Quantum Sensing & Metrology

Exploits the extreme sensitivity of quantum states to the environment to measure time, magnetic/electric fields, gravity, rotation and temperature with record precision. Examples: atomic clocks, NV-centre diamond magnetometers, atom interferometers (gravity gradiometers), SQUIDs for brain imaging (MEG), quantum LiDAR/radar, GPS-free inertial navigation.

Pillar	Core resource	Maturity
Computing	Superposition + entanglement	Early; prototypes with error-prone qubits
Communication	Photons, entanglement, no-cloning	QKD commercial; quantum internet = research
Sensing	Sensitivity of quantum states	Most mature; products already on market

9. National and Global Quantum Missions

Country / bloc	Programme	Highlights
India	National Quantum Mission (NQM), approved by Cabinet Apr 2023; ≈ ₹6,003 crore for 2023-24 to 2030-31; run by DST	Goals: build intermediate-scale quantum computers of 50-1000 physical qubits; satellite-based secure quantum communication over 2000 km within India; inter-city QKD over 2000 km; quantum sensing & metrology, new materials. Four Thematic Hubs (T-Hubs) at IISc Bengaluru (computing), IIT Madras (communication), IIT Bombay (sensing), IIT Delhi (materials). Startups such as QpiAI, and TIFR/ISRO/DRDO/RRI labs.
USA	National Quantum Initiative Act (2018); NQI Coordination Office; DOE & NSF quantum centres	Funding > US$1 billion/yr; players: IBM, Google, Microsoft, IonQ, Rigetti; NIST post-quantum cryptography standards (2024); Quantum-safe migration mandates.
European Union	Quantum Flagship (2018, €1 billion over 10 years); EU Quantum Strategy & Chips Act	Quantum computing, communication (EuroQCI network), sensing, simulation; strong national programmes (Germany, France, Netherlands); EuroHPC quantum computers.
China	National quantum priorities in Five-Year Plans; national lab at Hefei	Micius satellite (2016) - first satellite QKD & entanglement distribution over 1200 km; 2000+ km Beijing-Shanghai fibre backbone; Jiuzhang (photonic) and Zuchongzhi (superconducting) computers.
Others	UK National Quantum Technologies Programme; Canada, Australia, Japan, South Korea, Singapore, Finland	Combined public funding worldwide is now in the tens of billions of US dollars.

Exam tip / Remember

For India's NQM remember: 2023 - 2031, ₹6003 crore, 50-1000 qubits, 2000 km QKD, 4 T-Hubs. Compare briefly with the USA (NQI Act 2018), EU (Flagship 2018) and China (Micius satellite).

10. Solved Numerical Examples

Example 1 - Energy of a photon

Find the energy of a photon of green light, $\lambda = 500$ nm.

$E = hc/\lambda = (6.626\times10^{-34} \times 3\times10^{8}) / (500\times10^{-9}) = 3.98\times10^{-19}$ J ≈ 2.5 eV.

Example 2 - de Broglie wavelength of an electron

An electron moves at 10^6 m/s ($m = 9.1\times10^{-31}$ kg).

$\lambda = h/(mv) = 6.626\times10^{-34} / (9.1\times10^{-31} \times 10^6) = 7.3\times10^{-10}$ m ≈ 0.73 nm - similar to atomic spacing, so diffraction is observable.

Example 3 - Hydrogen transition

Electron falls from n = 3 to n = 2.

$\Delta E = -13.6(1/9 - 1/4) = 13.6 \times 0.1389 = 1.89$ eV → photon wavelength ≈ 656 nm (red H-alpha line).

Example 4 - Uncertainty principle

An electron is confined to $\Delta x = 1$ Å $= 10^{-10}$ m.

$\Delta p \geq \hbar/(2\Delta x) = 1.055\times10^{-34}/(2\times10^{-10}) = 5.3\times10^{-25}$ kg·m/s → $\Delta v \approx 5.8\times10^5$ m/s. Large! This is why electrons cannot be pinned inside atoms.

Example 5 - Probability from amplitudes

$|\psi\rangle = (\sqrt{3}/2)|0\rangle + (1/2)|1\rangle$. P(0) = 3/4 = 75%, P(1) = 1/4 = 25%. Check: 3/4 + 1/4 = 1 ✓.

11. Glossary of Key Terms

Term	Meaning
Quantum	Smallest discrete unit of a physical quantity (e.g. photon for light)
Wavefunction ψ	Mathematical object containing all information about a quantum system
Probability amplitude	Complex number whose squared magnitude gives probability
Eigenstate	State with a definite value for a measured quantity
Collapse	Sudden change of state to a definite outcome upon measurement
Interference	Adding of amplitudes giving bright/dark fringes
Planck constant ħ	6.626×10^{-34} J·s; sets the scale of quantum effects
Spin	Intrinsic angular momentum of a particle (no classical analogue)
Polarization	Orientation of oscillation of the electric field of light
Decoherence	Loss of quantum behaviour by interaction with environment
NISQ	Noisy Intermediate-Scale Quantum - today's imperfect devices
QKD	Quantum Key Distribution - secure sharing of cryptographic keys

12. Quick Revision & Likely Questions

- *Why did classical physics fail? Name 3 phenomena.*
- *State & explain the uncertainty principle with its significance.*
- *Explain superposition and entanglement with examples.*
- *Differentiate classical and quantum mechanics (table).*
- *What is quantization? Draw energy levels of hydrogen.*
- *Write a note on electrons, photons and atoms as quantum systems.*
- *Why is quantum technology strategically important? Explain 1st & 2nd quantum revolutions.*
- *Compare India's National Quantum Mission with those of the USA, EU and China.*

One-line summary

Quantum theory → discrete energies, wave-particle duality, probabilistic measurement → superposition & entanglement → new technologies in computing, communication and sensing → global race for leadership.